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Unit – 1 ⇝ Basic Matrix Algebra (Lecture Note)

Introduction

→ Matrix theory is a branch of mathematics that focuses on the study of matrices.

→ Matrices are one of the most powerful tools in mathematics. Matrix notation and operations are used in electronic spreadsheet programs for personal computer, which is used in different areas of business and science like budgeting, sales projection, cost estimation, analyzing the results of an experiment etc.

→ In graphic design, digital images are represented as matrices.

→ Matrices are also used in cryptography.

→ This mathematical tool is not only used in certain branches of sciences, but also in genetics, economics, sociology, modern psychology and industrial management.

Matrix

→ A rectangular array of **objects** arranged in rows and columns is known as matrix.

→ Objects can be numbers, expressions, symbols, etc.

→ These numbers, expressions or symbols are known as **element** or **entry** of matrix.

→ Matrix is denoted by upper case letters A, B, C, etc.

→ Following notations are used for matrices:

$$\begin{bmatrix} \quad \end{bmatrix}, \quad \begin{pmatrix} \quad \end{pmatrix} \quad \& \quad \begin{Vmatrix} \quad \end{Vmatrix}$$

- **Example:**

$$A = \begin{bmatrix} 1 & -2 \\ 3 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} + & - \\ \% & \cup \\ \Delta & \mathbb{N} \end{bmatrix}, \quad C = \begin{bmatrix} x+y & x^2+3 & x \\ 5 & y^2 & z \end{bmatrix}$$

→ In any matrix,

- Horizontal line of the elements is known as **row** of the matrix.
- Vertical line of the elements is known as **column** of the matrix.

→ In above examples, matrix A has 2 rows and 2 columns,

matrix B has 3 rows and 2 columns and

matrix C has 2 rows and 3 columns.

Order or Size of a Matrix

→ A matrix with **m** rows and **n** columns is known as a matrix of order **m × n**.

→ Matrix A of order m × n is written as $A_{m\times n}$.

→ In general, matrix X of order m × n can be written as follow:

$$X = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}_{m\times n}$$

Or

$$X = [\, a_{ij} \,]_{m\times n} \quad ; 1 \le i \le m, 1 \le j \le n$$

Where, a_{ij} is element of **i^{th}** row and **j^{th}** column of the matrix X.

→ The number of elements of matrix of order m × n is **m · n**.

Method 1 ⇝ Operations on Matrices and Types of Matrices

Operations on Matrices

→ Addition of Matrices

- Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two matrices with **equal order** $m \times n$, then addition of matrices A and B is denoted by $\mathbf{A + B}$ & defined as $A + B = [\mathbf{a_{ij}} + \mathbf{b_{ij}}]$.
- **Example**: Let $A = \begin{bmatrix} 1 & -2 & 3 \\ 6 & 9 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 8 & -5 \\ 0 & 2 & 11 \end{bmatrix}$, then

$$A + B = \begin{bmatrix} 1 & -2 & 3 \\ 6 & 9 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 8 & -5 \\ 0 & 2 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 6 & -2 \\ 6 & 11 & 13 \end{bmatrix}$$

→ Difference of Matrices

- Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two matrices with **equal order** $m \times n$, then difference of matrices A and B is denoted by $\mathbf{A - B}$ & defined as $A - B = [\mathbf{a_{ij}} - \mathbf{b_{ij}}]$.
- **Example:** Let $A = \begin{bmatrix} 1 & -2 & 3 \\ 6 & 9 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 8 & -5 \\ 0 & 2 & 11 \end{bmatrix}$, then

$$A - B = \begin{bmatrix} 1 & -2 & 3 \\ 6 & 9 & 2 \end{bmatrix} - \begin{bmatrix} 2 & 8 & -5 \\ 0 & 2 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -10 & 8 \\ 6 & 7 & -9 \end{bmatrix}$$

→ Multiplication of Matrix by a Scalar

- Let $A = [a_{ij}]$ be a matrix of order $m \times n$ and **k** be any **non-zero real number**, then multiplication of matrix A by a scalar k is written as $\mathbf{k \cdot A}$ or **kA** & defined as $k \cdot A = [\mathbf{k} \cdot \mathbf{a_{ij}}]$.

- **Example**: Let $A = \begin{bmatrix} 0 & 4 & -3 \\ -7 & 5 & 10 \end{bmatrix}$ and $k = 5$ be any non-zero real number, then

$$5A = 5 \cdot \begin{bmatrix} 0 & 4 & -3 \\ -7 & 5 & 10 \end{bmatrix} = \begin{bmatrix} 5 \cdot 0 & 5 \cdot 4 & 5 \cdot (-3) \\ 5 \cdot (-7) & 5 \cdot 5 & 5 \cdot 10 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 20 & -15 \\ -35 & 25 & 50 \end{bmatrix}$$

→ **Properties of Matrix Addition, Difference & Scalar Multiplication**

- Let A, B and C be three matrices with same order & α and β are scalars, then

(1) $A + B = B + A$ **(Commutative Law)**

(2) $A - B \neq B - A$

(3) $A + (B + C) = (A + B) + C$ **(Associative Law)**

(4) $A - (B - C) \neq (A - B) - C$

(5) $A + O = O + A = A$ **(Existence of Additive Identity)**

(6) $A + (-A) = (-A) + A = O$ **(Existence of Additive Inverse)**

(7) $\alpha(A \pm B) = \alpha A \pm \alpha B$ **(Distributive Law)**

(8) $(\alpha \pm \beta)A = \alpha A \pm \beta A$ **(Distributive Law)**

→ **Multiplication of Matrices**

- Multiplication of two matrices A and B is denoted as **AB** and it is defined if

No. of **columns** of A = No. of **rows** of B

- **Example**: Let $A = \begin{bmatrix} 1 & -2 & 3 \\ 6 & 9 & 2 \end{bmatrix}_{2\times3}$ and $B = \begin{bmatrix} 2 & 8 & 1 \\ 0 & 2 & 4 \\ -7 & -5 & 2 \end{bmatrix}_{3\times3}$

Here, No. of columns of matrix A = No. of rows of matrix B = 3

Therefore, AB is defined.

- Multiplication of two matrices B and A is denoted as **BA** and it is defined if

No. of **columns** of B = No. of **rows** of A

- **Example**: Let $A = \begin{bmatrix} 1 & -2 & 3 \\ 6 & 9 & 2 \end{bmatrix}_{2\times 3}$ and $B = \begin{bmatrix} 2 & 8 & 1 \\ 0 & 2 & 4 \\ -7 & -5 & 2 \end{bmatrix}_{3\times 3}$

  Here, No. of columns of matrix B = 3 and No. of rows of matrix A = 2.

  Therefore, BA is not defined.

- Let $A = [\,a_{ik}\,]$ be **m × p** matrix and $B = [\,b_{kj}\,]$ be a **p × n** matrix. The resultant matrix $AB = [\,c_{ij}\,]$ is **m × n** matrix. Here **ij^{th} − entry** is obtained by multiplying the **i^{th} row** of A by the **j^{th} column** of B.

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1p} \\ \vdots & \vdots & \cdots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ip} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mp} \end{bmatrix} \begin{bmatrix} b_{11} & \cdots & b_{1j} & \cdots & b_{1n} \\ b_{21} & \cdots & b_{2j} & \cdots & b_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_{p1} & \cdots & b_{pj} & \cdots & b_{pn} \end{bmatrix} = \begin{bmatrix} c_{11} & \cdots & c_{1n} \\ \vdots & \cdots & \vdots \\ \vdots & c_{ij} & \vdots \\ \vdots & \cdots & \vdots \\ c_{m1} & \cdots & c_{mn} \end{bmatrix}$$

- i. e., $c_{ij} = a_{i1} \cdot b_{1j} + a_{i2} \cdot b_{2j} + \cdots + a_{ip} \cdot b_{pj}$.

→ **Results on Matrix Multiplication**

- Let A, B and C are three matrices with suitable order & k be a scalar.

  (1) $(AB)C = A(BC)$ **(Associative Law)**

  (2) $A(B \pm C) = AB \pm AC$ **(Left Distributive Law)**

  (3) $(B \pm C)A = BA \pm CA$ **(Right Distributive Law)**

  (4) $IA = AI = A$ **(Existence of Multiplicative Identity)**

  (5) $k(AB) = (kA)B = A(kB)$

→ **Diagonal elements of a Matrix**

- In a matrix of order **n × n**, elements a_{ij} where **i = j** are known as **principal diagonal elements**, while elements a_{ij} where **i + j = n + 1** are known as **secondary diagonal elements.**

  i.e. Principal diagonal elements: $a_{11}, a_{22}, a_{33}, \ldots, a_{nn}$

  Secondary diagonal elements: $a_{1n}, a_{2(n-1)}, a_{3(n-2)}, \ldots, a_{n1}$

- Principal diagonal elements are also known as **diagonal elements** or **main diagonal elements.**
- Secondary diagonal elements are also known as **anti-diagonal elements.**
- **Example:**

$$A = \begin{bmatrix} \boxed{a_{11}} & a_{12} & a_{13} \\ a_{21} & \boxed{a_{22}} & a_{23} \\ a_{31} & a_{32} & \boxed{a_{33}} \end{bmatrix}$$

Principal Diagonal Elements

$$A = \begin{bmatrix} a_{11} & a_{12} & \boxed{a_{13}} \\ a_{21} & \boxed{a_{22}} & a_{23} \\ \boxed{a_{31}} & a_{32} & a_{33} \end{bmatrix}$$

Secondary Diagonal Elements

→ **Trace of a Matrix**

- In a matrix A of order **n × n**, the **sum** of all principal diagonal elements is known as **trace** of a matrix A.
- Trace of a matrix A is denoted by Trace(A) or **tr(A)**.
- **Example:** $A = \begin{bmatrix} \mathbf{9} & 2 & 4 \\ 0 & \mathbf{-1} & 1 \\ 0 & 5 & \mathbf{2} \end{bmatrix} \Rightarrow \text{tr}(A) = \mathbf{9} + (\mathbf{-1}) + \mathbf{2} = 10$

→ **Transpose of a Matrix**

- If we **interchange** all the rows of matrix A into columns or all the columns of matrix A into rows, then the resultant matrix is known as transpose of a matrix A.
- Transpose of a matrix A is denoted by $\mathbf{A^T}$ or $\mathbf{A'}$.
- If $A = [\, a_{ij} \,]$, then $A^T = [\, a_{ji} \,]$
- If order of matrix A is m × n, then the order of matrix A^T is **n × m**.
- **Example:** $A = \begin{bmatrix} 1 & 3 \\ 0 & -1 \\ -2 & 5 \end{bmatrix}_{3\times 2} \Rightarrow A^T = \begin{bmatrix} 1 & 0 & -2 \\ 3 & -1 & 5 \end{bmatrix}_{2\times 3}$

→ **Properties of Transpose of a Matrix**

(1) $(A \pm B)^T = A^T \pm B^T$

(2) $(AB)^T = B^T A^T$

(3) $(A^T)^T = A$

(4) $(k \cdot A)^T = k \cdot A^T$, k is a constant

Types of Matrices

→ **Row Matrix**

- A matrix with only **one row** is known as a row matrix.
- **Example:** $[\,1 \quad -2 \quad 5\,]$ is a row matrix of order 1×3.
- Order of row matrix is $\mathbf{1 \times n}$.

→ **Column Matrix**

- A matrix with only **one column** is known as a column matrix.
- **Example:** $\begin{bmatrix} 0 \\ \sqrt{5} \\ 3 \\ 1.3 \end{bmatrix}$ is a column matrix of order 4×1.
- Order of column matrix is $\mathbf{m \times 1}$.

→ **Square Matrix**

- A matrix is known as a square matrix, if the number of **rows** and **columns** are **equal**.
- **Example:** $\begin{bmatrix} 1 & 7 \\ 2 & 9 \end{bmatrix}$ is a square matrix of order 2×2.
- Square matrix A of order n, denoted by A_n.

→ **Diagonal Matrix**

- A **square** matrix in which all the elements **above** the principal diagonal & **below** the principal diagonal are **zero** is known as a diagonal matrix.

  i.e., $A = (a_{ij})_{n\times n}$ is diagonal matrix, if $a_{ij} = 0$ whenever $i \neq j$.
- A diagonal matrix A_n is denoted by $A_n = \text{diag}\,[\,a_{11},\ a_{22},\ \ldots,\ a_{nn}\,]$.
- **Example:**

$$A = [\,4\,], \qquad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad C = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$$= \text{diag}[\,4\,] \qquad = \text{diag}[\,0, 0\,], \qquad = \text{diag}[\,5, 0, -2\,]$$

→ **Scalar Matrix**

- A **diagonal** matrix in which all the principal diagonal elements are **same** is known as a scalar matrix.

  i.e., $A = (a_{ij})_{n\times n}$ is scalar matrix, if $a_{ij} = \begin{cases} 0, \text{whenever } i \neq j \\ k, \text{whenever } i = j \end{cases}$, where $k \in \mathbb{R}$

- **Example**: $A = [\,4\,]$, $B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, $C = \begin{bmatrix} \sqrt{5} & 0 & 0 \\ 0 & \sqrt{5} & 0 \\ 0 & 0 & \sqrt{5} \end{bmatrix}$

→ **Identity Matrix**

- A **diagonal** matrix in which all the principal diagonal elements are **1** is known as an identity matrix.

  i.e., $A = (a_{ij})_{n\times n}$ is identity matrix, if $a_{ij} = \begin{cases} 0, \text{whenever } i \neq j \\ 1, \text{whenever } i = j \end{cases}$

- Identity matrix of order n is denoted by $I_{n\times n}$ or $\mathbf{I_n}$ or I.

- **Example:** $I_1 = [\,1\,]$, $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

→ **Zero Matrix** OR **Null Matrix**

- A matrix in which all elements are **zero** is known as a zero matrix.
- Zero matrix of order n is denoted by $O_{n\times n}$ or O_n or **O**.
- **Example:** $O_1 = [\,0\,]$, $O_2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, $O_{3\times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$

→ **Upper Triangular Matrix**

- A **square** matrix in which all the elements **below** the principal diagonal are **zero** is known as an upper triangular matrix (UTM).

  i.e., $A = (a_{ij})_{n\times n}$ is an upper triangular matrix, if $\mathbf{a_{ij} = 0}$; $\forall$ **i, j** with **i > j**

- **Example:** $\begin{bmatrix} 9 & 2 & 4 \\ \mathbf{0} & 1 & 1 \\ \mathbf{0} & \mathbf{0} & 2 \end{bmatrix}$, $\begin{bmatrix} 1 & 2 & 2 & 4 \\ \mathbf{0} & 0 & -1 & 0 \\ \mathbf{0} & \mathbf{0} & 7 & 6 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & 9 \end{bmatrix}$

→ **Lower Triangular Matrix**

- A **square** matrix in which all the elements **above** the principal diagonal are **zero** is known as a lower triangular matrix (LTM).

  i.e., $A = (a_{ij})_{n\times n}$ is a lower triangular matrix, if $\mathbf{a_{ij} = 0}$; $\forall$ **i, j** with **i < j**

- **Example:** $\begin{bmatrix} 9 & \mathbf{0} & \mathbf{0} \\ 2 & 0 & \mathbf{0} \\ 6 & 8 & 2 \end{bmatrix}, \begin{bmatrix} 1 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ 2 & 1 & \mathbf{0} & \mathbf{0} \\ 3 & 0 & 7 & \mathbf{0} \\ 0 & 6 & 8 & 9 \end{bmatrix}$

→ **Real Matrix**

- If all the elements of matrix are **real numbers**, then such a matrix is known as a real matrix.

- **Example:** $A = \begin{bmatrix} \frac{1}{2} & \sqrt{5} \\ -3 & 0 \\ 1 & 4 \end{bmatrix}$

Special Matrices

→ **Symmetric Matrix**

- A **square** matrix $A = [\, a_{ij} \,]$ is known as a symmetric matrix, if $\mathbf{A^T = A}$.

  i.e., A square matrix $A = [\, a_{ij} \,]$ is a symmetric, if $\mathbf{a_{ij} = a_{ji}}$; $\forall$ i, j

- **Example:** Let $A = \begin{bmatrix} 9 & 2 & 6 \\ 2 & 1 & 8 \\ 6 & 8 & 2 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 9 & 2 & 6 \\ 2 & 1 & 8 \\ 6 & 8 & 2 \end{bmatrix}$

  $\Rightarrow A = A^T \Rightarrow$ A is a symmetric matrix.

→ **Skew-Symmetric Matrix or Anti-Symmetric Matrix**

- A **square** matrix $A = [\, a_{ij} \,]$ is known as a skew-symmetric matrix, if $\mathbf{A^T = -A}$.

  i.e., A square matrix $A = [\, a_{ij} \,]$ is a skew-symmetric matrix, if $\mathbf{a_{ij} = -\, a_{ji}}$; $\forall$ i, j

- **Principal diagonal** elements of a skew-symmetric matrix are always **zero** because we know that, for principal diagonal elements, **i = j**.

  Now, for a skew-symmetric matrix A, $a_{ij} = -\, a_{ji}$

  $\Rightarrow a_{ii} = -\, a_{ii}$ {taking i = j}

$\Rightarrow 2a_{ii} = 0$

$\Rightarrow a_{ii} = 0$

- **Example:** Let $A = \begin{bmatrix} 0 & -2 & 6 \\ 2 & 0 & 8 \\ -6 & -8 & 0 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 0 & 2 & -6 \\ -2 & 0 & -8 \\ 6 & 8 & 0 \end{bmatrix}$

$$\Rightarrow A^T = -\begin{bmatrix} 0 & -2 & 6 \\ 2 & 0 & 8 \\ -6 & -8 & 0 \end{bmatrix} = -A$$

$\Rightarrow A^T = -A \Rightarrow A$ is a skew-symmetric matrix.

→ **Remark**

- Every **square** matrix can be expressed as the **sum** of a **symmetric** and a **skew-symmetric** matrix as

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

Where, $P = \frac{1}{2}(A + A^T)$ is a symmetric matrix and $Q = \frac{1}{2}(A - A^T)$ is a skew-symmetric matrix.

- **Example:** Let $A = \begin{bmatrix} 2 & 4 & -2 \\ 6 & 0 & 8 \\ 2 & -4 & 4 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 2 & 6 & 2 \\ 4 & 0 & -4 \\ -2 & 8 & 4 \end{bmatrix}$

Then $\frac{1}{2}(A + A^T) = \begin{bmatrix} 2 & 5 & 0 \\ 5 & 0 & 2 \\ 0 & 2 & 4 \end{bmatrix}$ and

$$\frac{1}{2}(A - A^T) = \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & 6 \\ 2 & -6 & 0 \end{bmatrix}$$

where, $\frac{1}{2}(A + A^T)$ is a symmetric matrix and $\frac{1}{2}(A - A^T)$ is a skew-symmetric matrix and

$$\frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

$$= \begin{bmatrix} 2 & 5 & 0 \\ 5 & 0 & 2 \\ 0 & 2 & 4 \end{bmatrix} + \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & 6 \\ 2 & -6 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 4 & -2 \\ 6 & 0 & 8 \\ 2 & -4 & 4 \end{bmatrix} = A$$

- So, every **square** matrix can be expressed as the **sum** of a **symmetric** and a **skew-symmetric** matrix.

→ **Orthogonal Matrix**

- A square matrix A is known as an orthogonal matrix, if $\mathbf{AA^T = I = A^TA}$.
- **Example:** $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

Now, $AA^T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

$$= \begin{bmatrix} (0)(0)+(-1)(-1) & (0)(1)+(-1)(0) \\ (1)(0)+(0)(-1) & (1)(1)+(0)(0) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= I$$

Also, $A^TA = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} (0)(0)+(1)(1) & (0)(-1)+(1)(0) \\ (-1)(0)+(0)(1) & (-1)(-1)+(0)(0) \end{bmatrix}$$

$$= I$$

$\Rightarrow AA^T = I = A^TA \Rightarrow$ A is an orthogonal matrix.

→ **Idempotent Matrix**

- A square matrix A is known as an idempotent matrix, if $\mathbf{A^2 = A}$.
- **Example:** $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

Now, $A^2 = A \cdot A$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} (1)(1)+(0)(0) & (1)(0)+(0)(0) \\ (0)(1)+(0)(0) & (0)(0)+(0)(0) \end{bmatrix}$$

$$= \begin{bmatrix} 1+0 & 1+0 \\ 1+0 & 0+0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$= A$$

$\Rightarrow A^2 = A \Rightarrow A$ is an idempotent matrix.

→ **Involutory Matrix**

- A square matrix A is known as an involutory matrix, if $\mathbf{A^2 = I}$.
- **Example:** Let $A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

Now, $A^2 = A \cdot A$

$$= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} (-1)(-1)+(0)(0) & (-1)(0)+(0)(-1) \\ (0)(-1)+(-1)(0) & (0)(0)+(-1)(-1) \end{bmatrix}$$

$$= \begin{bmatrix} 1+0 & 0+0 \\ 0+0 & 0+1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\Rightarrow A^2 = I \Rightarrow A$ is an involutory matrix.

Method 2 ⇝ Determinant and Its Properties

Determinant of a Square Matrix

→ The determinant of a square matrix A is denoted by $|\mathbf{A}|$ or det(A) or Δ.

→ The determinant of matrix A = [a] of order $\mathbf{1 \times 1}$ is given by $|A| = \mathbf{a}$.

→ The determinant of matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ of order $\mathbf{2 \times 2}$ is given by $|A| = \mathbf{ad - bc}$.

→ The determinant of matrix $A = \begin{bmatrix} a & b & c \\ p & q & r \\ x & y & z \end{bmatrix}$ of order $\mathbf{3 \times 3}$ is given by:

$$|A| = a\begin{vmatrix} q & r \\ y & z \end{vmatrix} - b\begin{vmatrix} p & r \\ x & z \end{vmatrix} + c\begin{vmatrix} p & q \\ x & y \end{vmatrix}$$

Properties of Determinants

(1) Let A be a square matrix of order 'n' and 'k' be any non-zero scalar, then

$$\mathbf{|kA| = k^n |A|}$$

(2) Let A and B be square matrices of the equal order, then

$$\mathbf{|A||B| = |AB| = |B||A|}$$

(3) Let A be a square matrix of order n, then

$$\mathbf{|A^m| = |A|^m ; \forall\, m \in \mathbb{N}}$$

(4) Let A be a zero matrix of order n, then

$$\mathbf{|A| = 0}$$

(5) Let I be an identity matrix of order n, then

$$\mathbf{|I| = 1}$$

(6) The value of the determinant of a matrix and its transpose is same.

$$\mathbf{i.e.\ |A| = |A^T|}$$

(7) If each element in a row (or column) of a square matrix is zero, then the value of the determinant is zero.

(8) If any two rows (or columns) of a square matrix are identical (all corresponding elements are same), then value of determinant is zero.

(9) If any two rows (or columns) of a square matrix are interchanged, then sign of determinant changes by minus sign only.

(10) If the given matrix is diagonal matrix OR upper (lower) triangular matrix, then the value of the determinant is the **product of the diagonal elements**.

(11) If the **elements of a row** (or column) of a square matrix are expressed as the **sum** (or difference) of two or more terms, then the **determinant** can be expressed as the **sum** (or difference) of two or more determinants.

$$\text{i.e.,} \begin{vmatrix} a \pm d & b \pm e & c \pm f \\ p & q & r \\ x & y & z \end{vmatrix} = \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix} \pm \begin{vmatrix} d & e & f \\ p & q & r \\ x & y & z \end{vmatrix}$$

(12) The determinant of orthogonal matrix and involutory matrix is ± 1.

(13) The determinant of idempotent matrix is either **0** or **1**.

(14) The determinant of a **skew – symmetric** matrix of **odd order** is **zero**.

Elementary Row Operations

(1) **Interchange of any two rows**

- Notation: R_{ij} or $R_i \leftrightarrow R_j$; $i \neq j$.
- Meaning: **Interchange** of ith row and j^{th} row.
- **Example:**

$$A = \begin{bmatrix} 1 & -2 & 5 \\ 2 & 4 & 10 \\ 0 & 3 & -6 \end{bmatrix} \xrightarrow{\text{Applying } R_{13}} \sim \begin{bmatrix} 0 & 3 & -6 \\ 2 & 4 & 10 \\ 1 & -2 & 5 \end{bmatrix}$$

(2) **Multiplication of any row by a non-zero scalar k**

- Notation: $k \cdot R_i$
- Meaning: **Multiplication** of all the elements of **i^{th}** row by a non-zero scalar **k**
- **Example:**

$$A = \begin{bmatrix} 1 & -2 & 5 \\ 2 & 4 & 10 \\ 0 & 3 & -6 \end{bmatrix} \xrightarrow{\text{Applying } 3R_2} \sim \begin{bmatrix} 1 & -2 & 5 \\ 6 & 12 & 30 \\ 0 & 3 & -6 \end{bmatrix}$$

(3) **Add a multiple of one row to another row**

- Notation: $R_{ij}(k)$ or $R_j + k \cdot R_i$; $i \neq j$.
- Meaning: **Multiplication** of all the elements of ith row by a non-zero scalar **k** and **add** it into j^{th} row.

- **Example:**

$$A = \begin{bmatrix} 1 & -2 & 5 \\ 2 & 4 & 10 \\ 0 & 3 & -6 \end{bmatrix} \xrightarrow{\text{Applying } R_{12}(-2)} \sim \begin{bmatrix} 1 & -2 & 5 \\ 0 & 8 & 0 \\ 0 & 3 & -6 \end{bmatrix}$$

$$1 \quad -2 \quad 5 \times (-2) = -2 \quad 4 \quad -10$$

$$+ \quad 2 \quad 4 \quad 10$$

$$0 \quad 8 \quad 0$$

→ If we apply any elementary row operation on any matrix A, then the resultant matrix B is **equivalent** to the matrix A.

→ It is denoted by **B~A** and read as "B is equivalent to the matrix A".

→ **Remarks:**

(1) If we apply $\mathbf{R_{ij}}$ on a square matrix A, then the determinant of the resultant matrix is $-|\mathbf{A}|$.

(2) If we apply $\mathbf{k \cdot R_i}$ on a square matrix A, then the determinant of the resultant matrix is $\mathbf{k}\,|\mathbf{A}|$.

(3) If we apply $\mathbf{R_{ij}(k)}$ on a square matrix A, then the determinant of the resultant matrix remains **same**.

→ **Note:** The above determinant properties are equally valid for elementary column operations.

Examples of Method-2: Determinant and Its Properties

C	1	Evaluate the determinant of following matrices. $A = \begin{bmatrix} 6 & 7 \\ 5 & -3 \end{bmatrix}$, $B = \begin{bmatrix} 5 & -1 & 1 \\ 2\sqrt{2} & 3 & 0 \\ -2 & \sqrt{2} & 2 \end{bmatrix}$ **Answer:** $\mathbf{|A| = -53}$, $\mathbf{|B| = 40 + 4\sqrt{2}}$
C	2	(i) If A and B are square matrices of order 3 such that $\|A\| = -1$ and $\|B\| = 3$, then find the value of $\|3AB\|$. (ii) If A is a square matrix order 2 and $\|A\| = 3$, then find the value of $\|-5AA^T\|$. **Answer: (i) -81, (ii) 225**
C	3	If $Y = \begin{bmatrix} 1 & 2 & -7 \\ -2 & -3 & 0 \\ 5 & 4 & 2 \end{bmatrix}$, then show that $\|4Y\| = 64\|Y\|$.
C	4	If $A = \begin{bmatrix} x & 2 \\ 2 & x \end{bmatrix}$ and $\|A^4\| = 625$, then find the real value(s) of x. **Answer: $x = \pm 3$**
C	5	By using the properties of determinant and without expanding prove that $\begin{vmatrix} 1 & bc & a(b+c) \\ 1 & ca & b(c+a) \\ 1 & ab & c(a+b) \end{vmatrix} = 0.$
C	6	Without expanding, show that $\begin{vmatrix} 0 & b-a & c-a \\ a-b & 0 & c-b \\ a-c & b-c & 0 \end{vmatrix} = 0$

C	7	By using properties of determinant, show that $\begin{vmatrix} b+c & q+r & y+z \\ c+a & r+p & z+x \\ a+b & p+q & x+y \end{vmatrix} = 2\begin{vmatrix} a & p & x \\ b & q & y \\ c & r & z \end{vmatrix}$
C	8	By using the properties of determinant, show that $\begin{vmatrix} 1 & b & b^2 \\ 1 & c & c^2 \\ 1 & a & a^2 \end{vmatrix} = (a-b)(b-c)(c-a)$
C	9	Prove below statements. a. Determinant of orthogonal matrix is ± 1. b. Determinant of idempotent matrix is either 0 or 1. c. Determinant of involutory matrix is ± 1. d. Determinant of a skew – symmetric matrix of odd order is zero.

Method 3 ⇝ Echelon Forms and Rank of a Matrix

→ **Zero Row**

- A row in which all the elements are zero is known as zero row.
- **Example:** For $A = \begin{bmatrix} 1 & -2 & -6 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \\ 6 & -8 & 9 \end{bmatrix}$, second row is zero row.

→ **Leading/Pivot Element**

- The **first non-zero** element of any row is known as leading element of that row.
- **Example:** $A = \begin{bmatrix} 0 & \mathbf{-2} & 3 \\ \mathbf{1} & 0 & 4 \\ 0 & 0 & \mathbf{9} \end{bmatrix}$

  The leading element of 1st row is $\mathbf{-2}$,

  The leading element of 2nd row is **1**,

  The leading element of 3rd row is **9**.

→ **Order of a Row**

- Order of a row is the number of elements in that row **starting from the leading element.**
- **Example:**

$$A = \begin{bmatrix} 0 & \boxed{\mathbf{-2} & 5} \\ \boxed{\mathbf{1} & 0 & 4} \\ 0 & 0 & \boxed{\mathbf{6}} \end{bmatrix} \begin{matrix} \longrightarrow \text{Order} = 2 \\ \longrightarrow \text{Order} = 3 \\ \longrightarrow \text{Order} = 1 \end{matrix}$$

→ **Procedure to Convert the Matrix into Row - Echelon Form**

(1) Put all the **zero rows** at the **bottom** of the matrix.

(2) Arrange all the **rows** of the matrix in **decreasing** order.

(3) Select the **highest order row** whose **leading element** has some non-zero elements **below** it, and make them **zero** by using the leading element.

(4) Repeat the steps above.

→ **Note:** Row-Echelon form of a matrix is **not unique**.

- **Example:** Convert the following matrix into Row Echelon form.

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 6 & 2 & -4 & 3 \\ 0 & 4 & 1 & 0 \\ 0 & 5 & 0 & 1 \end{bmatrix}$$

Step (1): Put all zero rows at the bottom of the matrix.

Here, in given matrix A, first row is a zero row, which must be the last row of matrix A.

$$\overset{R_{14}}{\sim} \begin{bmatrix} 0 & 5 & 0 & 1 \\ 6 & 2 & -4 & 3 \\ 0 & 4 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Step (2): Arrange all the rows in decreasing order.

$$\overset{R_{12}}{\sim} \begin{bmatrix} 6 & 2 & -4 & 3 \\ 0 & 5 & 0 & 1 \\ 0 & 4 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Step (3) Select the highest order row whose leading element has some non-zero elements below it, and make them zero by using the leading element.

As all the elements below the leading element of 1st row are zero.

So, select second row.

$$\overset{R_{23}\left(-\frac{4}{5}\right)}{\sim} \begin{bmatrix} 6 & 2 & -4 & 3 \\ 0 & 5 & 0 & 1 \\ 0 & 0 & 1 & -\frac{4}{5} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Step (4): Repeat Step (2) and Step (3) for **3rd row**.

As matrix after above operation is in decreasing order, let us make all the elements zero below the leading element of 3rd row.

As all the elements below the leading element 1 of 3rd row is zero, there is nothing to do.

⇒ The row echelon of given matrix A is

$$\sim \begin{bmatrix} 6 & 2 & -4 & 3 \\ 0 & 5 & 0 & 1 \\ 0 & 0 & 1 & -\frac{4}{5} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

→ **Procedure to Convert the Matrix into Reduced Row - Echelon Form**

(1) Put all the **zero rows** at the **bottom** of the matrix.

(2) Arrange all the **rows** of the matrix in **decreasing** order.

(3) Select the top row of **highest order** and make its leading element **1, if not**.

(4) By using the leading element of the selected row, make all entries above and below it **equal to 0**.

(5) Repeat the steps for the **next higher order** row.

→ **Note:** Reduced row echelon form of matrix is **unique**.

- **Example:** Convert the following matrix into Reduced Row Echelon form.

$$A = \begin{bmatrix} 0 & 2 & 2 & 6 & 4 \\ 0 & 1 & 0 & 4 & 2 \\ 0 & 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 1 & 3 \end{bmatrix}$$

Step (1): Put all zero rows at the bottom of the matrix.

Since, there is no zero row in given matrix, so we have nothing to do.

Step (2): Arrange all rows of the matrix in decreasing order.

$$\sim A = \begin{bmatrix} 0 & 2 & 2 & 6 & 4 \\ 0 & 1 & 0 & 4 & 2 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix} \quad R_{34}$$

Step (3): Select the top row of highest order and make its leading element 1.

$$\left(\frac{1}{2}\right)\mathbf{R_1}$$

$$\sim A = \begin{bmatrix} 0 & 1 & 1 & 3 & 2 \\ 0 & 1 & 0 & 4 & 2 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

Step (4): By using the leading element of the selected row, make all entries above and below it equal to 0.

$$\mathbf{R_{12}(-1)}$$

$$\sim A = \begin{bmatrix} 0 & 1 & 1 & 3 & 2 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

Step (5): Repeat the steps for the **next higher order** row.

Leading element of 2nd row is -1, so make it 1.

$$\mathbf{(-1)R_2}$$

$$\sim A = \begin{bmatrix} 0 & 1 & 1 & 3 & 2 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

With the help of leading element of 2nd row, make all the elements zero **above** & **below** the leading element of 2nd row.

$$\mathbf{R_{21}(-1)}$$

$$\sim A = \begin{bmatrix} 0 & 1 & 0 & 4 & 2 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

Leading element of 3rd row is already 1, so make all the elements zero **above** & **below** the leading element of 3rd row.

$$\mathbf{R_{31}(-4)}, \qquad \mathbf{R_{32}(1)}$$

$$\sim A = \begin{bmatrix} 0 & 1 & 0 & 0 & -10 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

Leading element of 4th row is 4, so make it 1.

$$\left(\frac{1}{4}\right) R_4$$

$$\sim A = \begin{bmatrix} 0 & 1 & 0 & 0 & -10 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

With the help of leading element of 4th row, make all the elements zero above the leading element of 4th row.

$$R_{43}(-3), \quad R_{42}(-3), \quad R_{41}(10)$$

$$\sim A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Which is reduced row echelon form of matrix A.

→ **Difference between Row-Echelon and reduce Row-Echelon Form**

Row-Echelon Form	Reduce Row-Echelon Form
The leading entries can be anything.	The leading entries must be 1(one).
Make all the entries zero below the leading element of each row.	Make all the entries zero below and above leading element of each row.
Row echelon form is not unique.	Reduced Row echelon form is unique.

Minor of a Matrix

Let A be m × n matrix.

- Determinant of a **square submatrix** of a given matrix A is known as **minor** of A.
- If the square submatrix is of order r, then it's **minor is also of order r**.

Rank of a Matrix

→ Let A be an m × n matrix. Then the non-negative integer **r** is known as the **rank** of A if

(1) There is **at least one** minor of order **r**, which is non-zero and,

(2) Every minor of order **(r + 1)** is zero.

→ Rank of a matrix A is denoted by **ρ(A)** or **Rank(A)**. (Read ρ as rho)

→ The **rank** of a matrix is equal to the number of non-zero rows in its **row echelon form**.

→ **Notes:**

(1) If A be an m × n matrix, then $\rho(A) \leq \min\{m, n\}$

(2) $\rho(A) = \rho(A^T)$

→ **Procedure to Find the Rank of a Matrix by Minor/Determinant Method**

(1) Let A be a non-zero m × n matrix, where m ≤ n

Since, $\rho(A) \leq \min\{m, n\}$

$\Rightarrow \boldsymbol{\rho(A) \leq m}$

(2) If we cannot find at least one minor of order m (i.e., all the minors of order m are zero), then ρ(A) can never be m.

$\Rightarrow \boldsymbol{\rho(A) < m}$

- In this case, if we can find at least one minor of order $m - 1$, which is non-zero, then $\rho(A) = m - 1$
- Otherwise, $\rho(A) < m - 1$

(3) Do the above process until we get at least one non-zero minor.

- **Example:**

Find the rank of a matrix $A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 0 \\ 7 & 8 & 9 & 0 \end{bmatrix}$ by minor method.

Solution: Since the matrix B is of order 3 × 4.

$\rho(A) \leq \min\{3,4\}$

$\Rightarrow \rho(A) \leq 3$

→ Now, try to find at least one minor of order 3, which is non-zero.

$$M_1 = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 0 \qquad M_2 = \begin{vmatrix} 1 & 2 & 0 \\ 4 & 5 & 0 \\ 7 & 8 & 0 \end{vmatrix} = 0$$

$$M_3 = \begin{vmatrix} 1 & 3 & 0 \\ 4 & 6 & 0 \\ 7 & 9 & 0 \end{vmatrix} = 0 \qquad M_4 = \begin{vmatrix} 2 & 3 & 0 \\ 5 & 6 & 0 \\ 8 & 9 & 0 \end{vmatrix} = 0$$

Since, all the minors of order 3 are zero, $\rho(A) \leq 2$.

→ Now, try to find at least one minor of order 2, which is non-zero.

$$M_5 = \begin{vmatrix} 1 & 0 \\ 7 & 0 \end{vmatrix} = 0 - 0 = 0$$

$$M_6 = \begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix} = 5 - 8 = -3 \neq 0$$

$\Rightarrow \rho(A) = 2$

Examples of Method-3: Echelon Forms and Rank of a Matrix

C	1	Convert the matrix $\begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & -4 & 5 & 2 \\ 2 & 3 & 1 & 6 \end{bmatrix}$ into row–echelon form. **Answer:** $\begin{bmatrix} \mathbf{1} & \mathbf{2} & \mathbf{0} & \mathbf{3} \\ \mathbf{0} & \mathbf{-1} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{2} \end{bmatrix}$
C	2	Convert the matrix $\begin{bmatrix} 2 & 4 & 3 & 4 \\ 1 & 2 & 1 & 3 \\ 1 & 3 & 2 & 2 \\ 3 & 7 & 4 & 8 \end{bmatrix}$ into reduced row-echelon form. **Answer:** $\begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{3} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} & \mathbf{1} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} & \mathbf{-2} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}$
C	3	Determine whether the following matrices are in either row echelon form or reduced row echelon form or both or none.

		$A = \begin{bmatrix} 0 & 2 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 8 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 4 & 6 & 5 \\ 6 & 0 & 8 & 3 \\ 0 & 0 & 5 & 7 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 6 \\ 0 & 5 \\ 0 & 3 \end{bmatrix}$, $D = \begin{bmatrix} 0 & 4 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$, $E = \begin{bmatrix} 5 & -4 & 6 & 5 \\ 0 & 0 & 8 & 3 \\ 0 & 0 & 0 & 7 \end{bmatrix}$, $F = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $G = [\,6\,]$, $H = [\,3 \quad 2\,]$, $I = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$, $J = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$, $K = [\,1 \quad 2\,]$. **Answer: A → None, B → None, C → None, D → REF, E → REF, F → REF, G → REF, H → REF, I → Both, J → Both, K → Both**
C	4	Define the rank of a matrix. Compute the rank of the matrix $A = \begin{bmatrix} 1 & 5 & 3 & -2 \\ 2 & 0 & 4 & 1 \\ 4 & 8 & 9 & -1 \end{bmatrix}$. **Answer: $\rho(A) = 3$**
C	5	Find the rank of matrix $A = \begin{bmatrix} 1 & -1 & 2 & 0 \\ 3 & 1 & 0 & 0 \\ -1 & 2 & 4 & 0 \end{bmatrix}$ by minor method. **Answer: $\rho(A) = 3$**

Linear Equation

→ In two dimensions, a line in a rectangular xy-coordinate system can be represented by an equation of the form $ax + by = c$ (a, b not both 0)

→ In three dimensions, a plane in a rectangular xyz-coordinate system can be represented by an equation of the form $ax + by + cz = d$ (a, b, c not all 0).

→ More generally, a linear equation in n variables $x_1, x_2, \ldots, x_n$ can be expressed in the form

$$a_1x_1 + a_2x_2 + a_3x_3 + \cdots + a_nx_n = b$$

where $a_1, a_2, \ldots, a_n$ and b are constants, and the a_i's are not all zero.

→ Linear equation **does not** involve any products or roots of variables and no variables involved in trigonometric, exponential, or logarithmic functions.

Also, the **power** of all variables is **one**.

- **Examples:**

(1) $3x - 2y = 7$

(2) $\frac{1}{2}x - \pi y = \sqrt{2}$

→ Non-linear equation does involve any products or roots of variables and variables involved in trigonometric, exponential, or logarithmic functions.

- **Examples:**

(1) $xy + z = 7$

(2) $e^x - 2y = 4$

System of Simultaneous Linear Equations

→ A **finite set** of linear equations is known as **system of linear equations** or, more briefly, a **linear system**. The variables are also known as unknowns.

→ A general linear system of m equations and n variables $x_1, x_2, \ldots, x_n$ can be written as:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n = b_m$$

→ Matrix representation of the above system is $AX = B$, where

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}_{m\times n}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1} \text{ and } B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}$$

→ For the above system,

- Matrix A is known as co-efficient matrix.
- Matrix X is known as variable matrix.
- Matrix B is known as constant matrix.

→ **Augmented Matrix**

- The augmented matrix of the above system is denoted as **[A | B]** or **[A : B]** and defined as

$$[A \mid B] = \left[\begin{array}{ccccc|c} a_{11} & a_{12} & a_{13} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} & b_2 \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} & b_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} & b_m \end{array}\right]_{m\times(n+1)}$$

→ Every system of linear equations has either

- Unique solution (exactly one solution) or
- Infinitely many solutions or
- No solution.

→ **Consistent System**

- If a system of linear equations has **at least one solution**, then that system is known as consistent system.

→ **Inconsistent System**

- If a system of linear equations has **no solution**, then that system is known as inconsistent system.

Homogeneous System of Linear Equations

→ In a system AX = B, if constant matrix B is zero matrix (O), then the system **AX = O** is known as **Homogeneous system** of linear equations.

→ Let us consider the system of Homogeneous equation

$$x_1 - 2x_2 + 3x_3 = 0$$

$$2x_1 - x_2 + 3x_3 = 0$$

$$-x_1 + x_2 - 2x_3 = 0$$

→ Put, $x_1 = 0$, $x_2 = 0$ and $x_3 = 0$ in above system

So, Solution $(x_1, x_2, x_3) = (0, 0, 0)$ is **trivial solution.**

→ Homogeneous system always has a trivial solution, so the homogeneous system is always **consistent**.

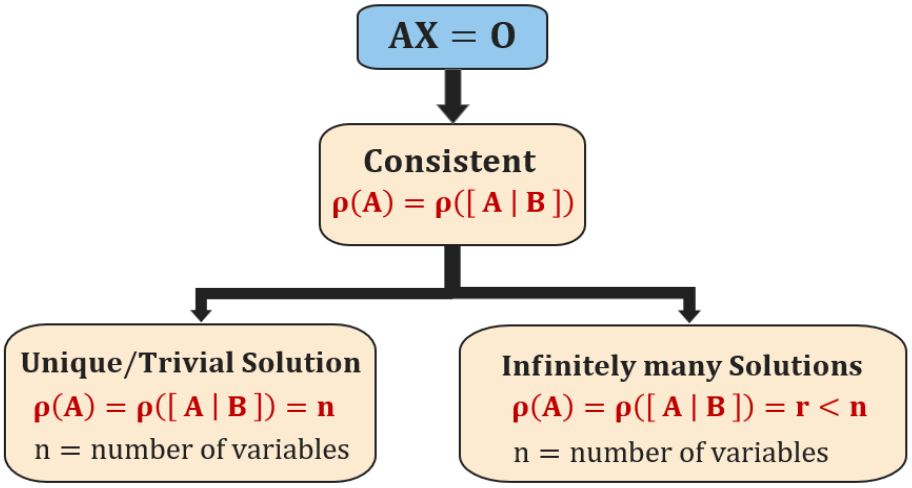
→ In case of infinitely many solutions,

Number of free variables $= \mathbf{n - \rho(A)}$;

where, free variable is such variable that can take any value in solution set.

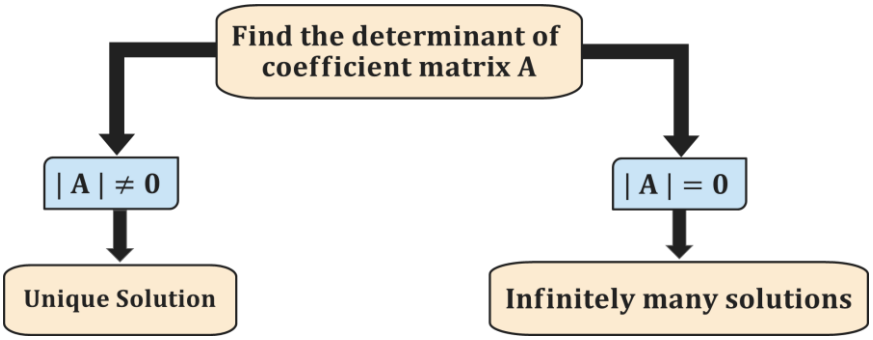
- n is number of variables
- ρ(A) is rank of coefficient matrix A.

→ **Idea of solution by Rank Method for Homogeneous System**



→ **Idea of solution by Determinant Method for Homogeneous System**

→ For a homogeneous system, in which number of equations and number of variables are equal.

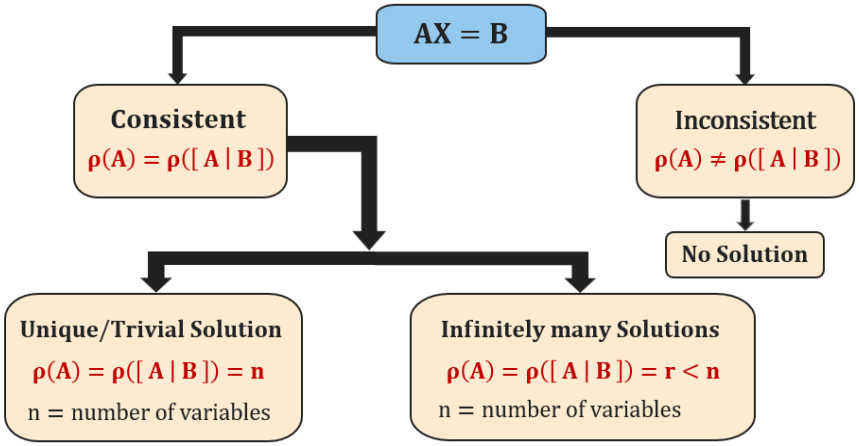


Non-homogeneous System of Linear Equations

→ In a system AX = B, if constant matrix B is not a zero matrix, then the system **AX = B** is known as **non-homogeneous system** of linear equations.

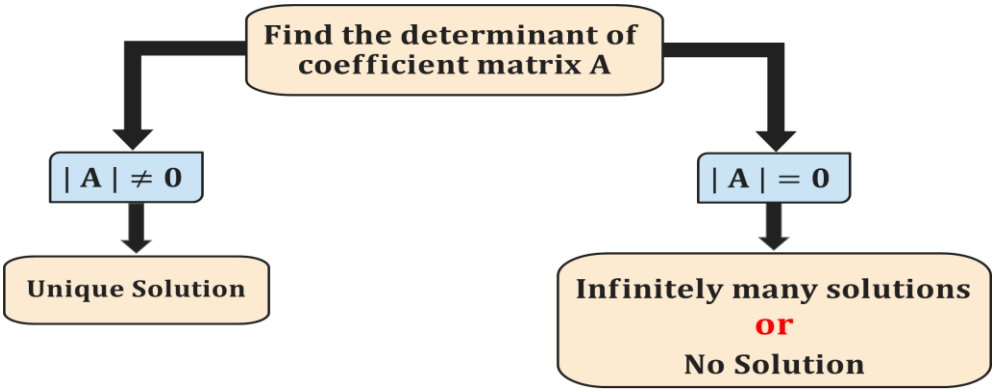
→ Non-homogeneous system has either unique solution, infinitely many solutions or no solution.

→ **Idea of solution by Rank Method for Non-Homogeneous System**



→ **Idea of solution by Determinant Method for Non-Homogeneous System**

→ For a non-homogeneous system, in which number of equations and number of variables are equal.



Method 4 ⇝ Gauss Elimination Method

→ **Procedure to solve the system of simultaneous linear equations by using the Gauss – Jordan Method:**

(1) Convert the given system of linear equations into matrix form as **AX = B**, where A is coefficient matrix, X is variable matrix and B is constant matrix.

(2) Convert augmented matrix **[A | B]** into **row echelon form**.

(3) Apply **back substitution** to get equations.

(4) **Solve** the equations to find the unknown variables.

Examples of Method-4: Gauss Elimination Method

C	1	By using Gauss Elimination Method solve the following system: $-x + 3y + 4z = 30, \quad 3x + 2y - z = 9, \quad 2x - y + 2z = 10$ **Answer**: $(\mathbf{2},\ \mathbf{4},\ \mathbf{5})$
C	2	Solve the following system by using Gauss Elimination Method: $-\frac{1}{x} + \frac{3}{y} + \frac{4}{z} = 30, \quad \frac{3}{x} + \frac{2}{y} - \frac{1}{z} = 9, \quad \frac{2}{x} - \frac{1}{y} + \frac{2}{z} = 10$ **Answer**: $\left(\mathbf{\frac{1}{2}},\ \mathbf{\frac{1}{4}},\ \mathbf{\frac{1}{5}}\right)$
C	3	Solve by using Gauss Elimination Method: $-2b + 3c = 1, \quad 3a + 6b - 3c = -2, \quad 6a + 6b + 3c = 5$ **Answer**: **The given system has no soluton**
C	4	Find the solution set of the given system of linear equations by using Gauss Elimination Method. $x_1 + 3x_2 - 2x_3 + 2x_5 = 0, \quad 2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 = -1,$ $5x_3 + 10x_4 + 15x_6 = 5, \quad 2x_1 + 6x_2 + 8x_4 + 4x_5 + 18x_6 = 6.$ **Answer**: $\left\{\left(\mathbf{-4s - 3t - 2r},\ \mathbf{t},\ \mathbf{-2s},\ \mathbf{s},\ \mathbf{r},\ \mathbf{\frac{1}{3}}\right) \mid \mathbf{r},\ \mathbf{s},\ \mathbf{t} \in \mathbb{R}\right\}$

C	5	The augmented matrix of a linear system has the form for $a, b, c \in \mathbb{R}$ is $\left[\begin{array}{ccc:c} -2 & 3 & 1 & a \\ 1 & 1 & -1 & b \\ 0 & 5 & -1 & c \end{array}\right]$. 1) Determine when the linear system is consistent. 2) Determine when the linear system is inconsistent. 3) Does the linear system have a unique solution or infinitely many solutions? **Answer: 1) $c - a - 2b = 0$, 2) $c - a - 2b \neq 0$,** **3) Infinitely many solutions**
C	6	For which values of λ & $k \in \mathbb{R}$, the following system has (i) No solution, (ii) Unique solution and (iii) Infinitely many solutions? $x + y + z = 6, \quad x + 2y + 3z = 10, \quad x + 2y + \lambda z = k$ **Answer: (i) $\lambda = 3$, $k \neq 10$ (or $k \in \mathbb{R} \setminus \{10\}$) ⇝ No solution** **(ii) $\lambda \neq 3$, $k \in \mathbb{R}$ ⇝ Unique solution** **(iii) $\lambda = 3$, $k = 10$ ⇝ Infinitely many solutions.**
C	7	Find the real value of λ for which given equations have solution other than $x = y = z = 0$. Also, find the solution for each real value of λ. $(1 - \lambda)x - y + z = 0, \quad 2x + (1 - \lambda)y = 0, \quad 2y - (1 + \lambda)z = 0$ **Answer: For $\lambda = 1$, the given system has solution other than $x = y = z = 0$.** **Solution: $(x, y, z) = (0, k, k); k \in \mathbb{R}$**

Method 5 ⇝ Gauss – Jordan Elimination Method

→ **Procedure to solve the system of simultaneous linear equations by using the Gauss – Jordan Elimination Method:**

(1) Convert the given system of linear equations into matrix form as **AX = B**,

where A is coefficient matrix, X is variable matrix and B is constant matrix.

(2) Convert augmented matrix **[A | B]** into **reduced row echelon form.**

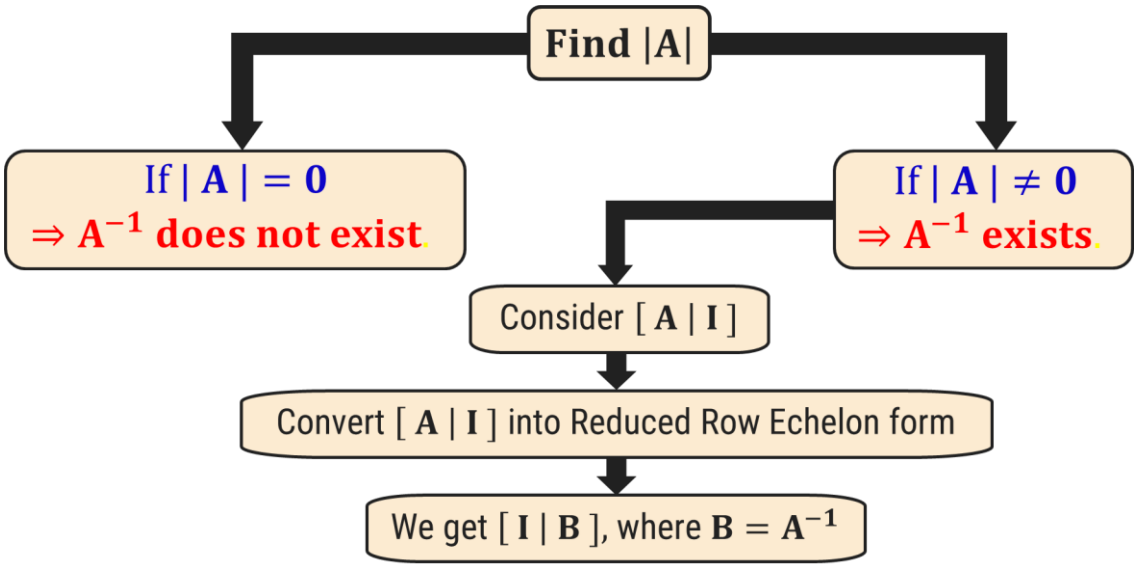
(3) Convert matrix form into linear system to get the value of unknown variables.

Examples of Method-5: Gauss - Jordan Elimination Method

C	1	Solve the system of linear equations by using Gauss – Jordan Elimination Method. $x + y + 2z = 8, \quad -x - 2y + 3z = 1, \quad 3x - 7y + 4z = 10$ **Answer**: **(3, 1, 2)**
C	2	Find the solution set of following system by using Gauss – Jordan Elimination Method. $2x_1 + 2x_2 - x_3 + x_5 = 0, \quad -x_1 - x_2 + 2x_3 - 3x_4 + x_5 = 0,$ $x_1 + x_2 - 2x_3 - x_5 = 0, \quad x_3 + x_4 + x_5 = 0$ **Answer**: $\{(-s, -t, s, -t, 0, t) \mid s, t \in \mathbb{R}\}$

Method 6 ⇝ Inverse of a Matrix by the Gauss – Jordan Method

→ **Procedure to find Inverse of a Matrix by the Gauss – Jordan Method**



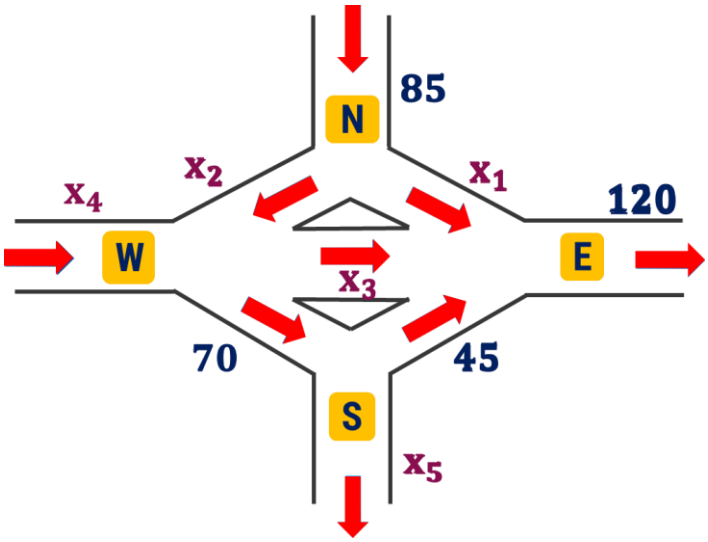
Examples of Method-6: Inverse of a Matrix by the Gauss – Jordan Method

C	1	Find the inverse of the given matrix by Gauss-Jordan Method, if possible. $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & 1 & 0 \end{bmatrix}$. **Answer: A^{-1} does not exists because $\lvert A \rvert = 0$**
C	2	Find the inverse of matrix $A = \begin{bmatrix} 2 & 1 & 3 \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ by Gauss-Jordan Method. **Answer: $\frac{1}{6}\begin{bmatrix} -1 & 3 & -1 \\ -7 & 3 & 5 \\ 5 & -3 & -1 \end{bmatrix}$**

Method 7 ⇝ Applications of Linear Systems

→ Linear systems are very useful to model real life problems. There are various applications of linear system which can be used to model mechanical systems, electrical networks, Chemical equations etc.

Examples of Method-7: Applications of Linear Systems

C	1	Predicting the flow of traffic at $x_1, x_2, \ldots \ldots x_5$ for the following network of streets and solve the system of linear equations.  **Answer:** $x_1 = 75 - t$, $x_2 = 10 + t$, $x_3 = t$, $x_4 = 60$, $x_5 = 25$; $0 \leq t \leq 75$ and $t \in \mathbb{N}$.
C	2	An investor has ₹1,20,000 to invest in three types of bonds. Short term, mid-term and long term. Short term pays 4% annually, mid-term pays 5% and long term pay 10% . He has invested equal amount in short term and mid term bonds. How much should he invest in each type to get a total annual income of 6% with equal amounts invested in all types of bonds? **Answer:** (43636.38, 43636.38, 32727.27)

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